

3. OVULATION, OVARIAN AND UTERINE CYCLES, HORMONAL REGULATION

The release of the female gamete is ovulation. It occurs in the middle of the female cycle, on the 14th day of the menstrual cycle. The entire cumulus has then detached from the rest of the granulosa (under the action of proteolytic enzymes) and floats in the follicular cavity. The follicle wall remains in the ovary and constitutes the dehiscent follicle.

1. MECHANISMS OF OVULATION

- the mature follicle protrudes from the surface of the ovary.
- At the level of the theca interna, vasodilation causes an increase in the volume of the follicular cavity. After collapse of the Slavjanski membrane, the oocyte surrounded by follicle cells floats in the follicular fluid.
- In the apex region, vasoconstriction and necrosis of the ovarian wall occur. The oocyte completes its cytoplasmic and nuclear maturation which results in the formation of oocyte II which will be blocked in metaphase of meiosis II.
- At the time of ovulation, the mobile auricle of the fallopian tube covers the ovary. It collects oocyte II. Proper mobility of the auricle is essential for reproductive function. The oocyte quickly migrates to the tubal ampulla where fertilization takes place. Migration is passive. It is facilitated by the fluid flow from the auricle to the uterus and the movements of the cilia of the tubal epithelium.

The oocyte is a spherical cell of 150 µm in diameter, relatively inert, surrounded by its envelopes

2. FORMATION OF THE CORPUS LUTEUM

Immediately after ovulation, the follicle transforms into a temporary endocrine gland, the corpus luteum (luteinization). The granulosa and theca interna cells modify their hormone synthesis and become luteal and paraluteal cells, producing progesterone and estrogen.

- If there is no implantation, the corpus luteum regresses (luteolysis). This is the cyclic corpus luteum or progestin body. After the involution of the corpus luteum, a small mass of fibrous tissue, the white body or corpus albicans, remains in the ovary.

- If there is a pregnancy, the corpus luteum persists until the 3rd month. This is the corpus luteum of pregnancy or gestative body. Its function is to produce progesterone to maintain the pregnancy.

3. SEXUAL CYCLES

The cyclical functioning of the female reproductive system begins at puberty and ends at menopause. It involves the ovaries and the uterus.

3.1. Ovarian cycle

There are two phases separated by ovulation, which corresponds to the expulsion of an oocyte by a mature follicle:

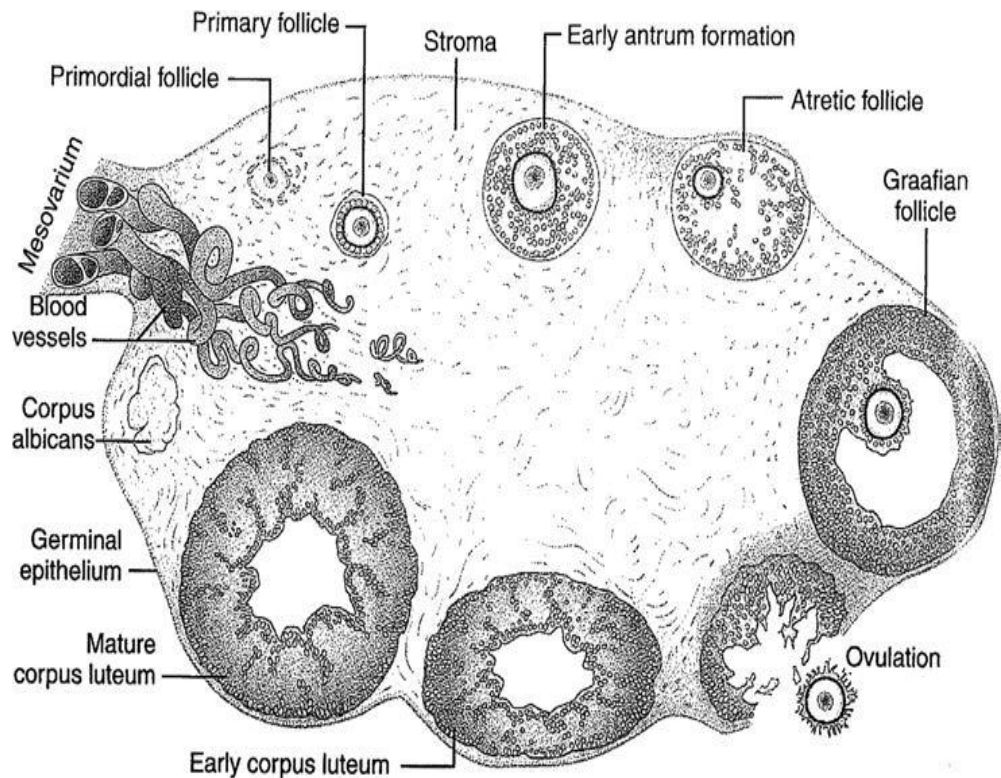


Figure 1. Ovulation and formation of corpus luteum

3.1.1. Follicular phase

Lasting between 12 and 18 days, it corresponds to the maturation of one of the follicles while others which had begun to grow during previous cycles degenerate (atresia).

3.1.2. Luteal phase

Of a constant duration of 13 to 14 days marked by the transformation of the remains of the follicle into a corpus luteum which degenerates in the absence of fertilization.

NB: The cells of the cervix also undergo changes: they release mucus, the mucus, which evolves during the uterine cycle: this cycle of cervical mucus is characterized by secretions presenting a mesh of mucus; very tight in the follicular phase, it becomes loose during the ovulatory period, facilitating the passage of spermatozoa towards the uterine cavity.

3.2. Uterine cycle (menstrual cycle)

It results in particular in the thickening of the uterine lining, the endometrium thus prepares itself for a possible implantation of the embryo. This cycle lasts on average 28 days and begins with the onset of menstruation. During the woman's menstrual cycle, profound changes appear, this evolution is reflected in several phases:

3.2.1. Proliferative phase

Significant thickening of the endometrium, lasting one or two weeks, during the follicular phase.

3.2.2. Secretory phase

Significant development of tube-like glands in the post-ovulatory period, there is formation of uterine lace (=structure hollowed out by glands and richly vascularized).

3.2.3. Menstrual phase

At the end of the cycle, if there has been no fertilization, the spiral arteries of the mucosa dilate and their walls rupture, the mucosa detaches and is eliminated in a blood flow: these are menstruations. This is the first day of a new cycle.

3.3. Synchronism of ovarian and uterine cycles

The ovaries have a dual function: on the one hand they produce reproductive cells, on the other hand they secrete sex hormones (estrogen and progesterone).

During the sexual cycle, certain ovarian structures release hormones that allow the ovary and uterus to function in synchrony. In the follicular phase, these are the granulosa cells and the internal theca, and in the luteal phase, the corpus luteum.

During the follicular phase, the cells of the internal theca secrete estrogen, the amount secreted increases with the growth of the follicle, it decreases slightly at the time of ovulation.

In the luteal phase, the corpus luteum cells (luteal cells) secrete estrogen and progesterone.

Ovarian hormones are released cyclically, with estradiol predominating in the follicular phase with a peak just before ovulation; the two hormones act together on the uterine lining in the luteal phase.

4. HORMONAL CONTROL

Hormonal regulation in women is carried out as follows:

The adenohypophysis, stimulated by GnRH, releases FSH and LH. FSH and LH cause changes in the ovary, which in turn produces estrogen and progesterone.

- **During the pre-ovulatory phase : folliculo-estrogenic phase**

The anterior pituitary gland initially secretes FSH, which stimulates follicular development: multiplication of granulosa cells with development of the zona pellucida. The anterior pituitary gland begins to secrete LH in progressively increasing quantities, and in the final stages of follicular maturation, LH stimulates the differentiation of the theca interna cells, which in turn secrete estrogen. FSH and LH work synergistically: neither FSH nor LH, acting in isolation, can cause follicular growth.

- **Around the middle of the cycle: ovulation**

Gonadotropins cause the dissociation of cumulus oophorus cells, the resumption of the first division of meiosis, which is completed within a few hours. Thirty-six hours after the peak,

oocyte II is released. Ovulation is due to a mid-cyclical discharge of LH and FSH or gonadotropic discharge [pituitary surge of LH and FSH].

- **During the post-ovulatory phase : luteal phase**

After ovulation, the remains of the follicle change under the influence of LH (which continues to be secreted by the pituitary gland) into an endocrine structure: the corpus luteum, which begins to secrete mainly progesterone and little estrogen.

4.1. Cyclical variations

- For pituitary hormones :FSH and LH have a variable rate during the cycle:
 - FSH is high early in the cycle and has a pre-ovulatory peak in mid-cycle.
 - LH has a low level throughout the cycle and also a very significant pre-ovulatory peak in the middle of the cycle.
- For ovarian hormones: estrogen, low at the beginning of the cycle, their level rises and there is a significant peak 12 to 24 hours before ovulation. In the luteal phase, there is a slight increase in estrogen followed by a decrease at the time of menstruation.
- Progesterone levels vary throughout the menstrual cycle: they are very low throughout the pre-ovulatory phase, rise after ovulation until the 8th day of the luteal phase, and then decline until menstruation.

4.2. Roles of ovarian hormones

These steroid sex hormones determine the appearance of primary sexual characteristics in the fetus, the appearance of secondary sexual characteristics at puberty and control oogenesis.

4.2.1. Estrogens

- They influence the cyclical variations of the epithelium of the fallopian tube, the endometrium: proliferation of the endometrial glands, reconstitution of the uterine mucosa.
- Stimulate contractions of the myometrium.
- Cause cervical mucus to be abundant at the time of fertilization.

4.2.3. Progesterone

- Prepares the uterine lining for implantation: growth of glands and secretion.
- Inhibits contractions of the myometrium.
- Changes the abundance of mucus.

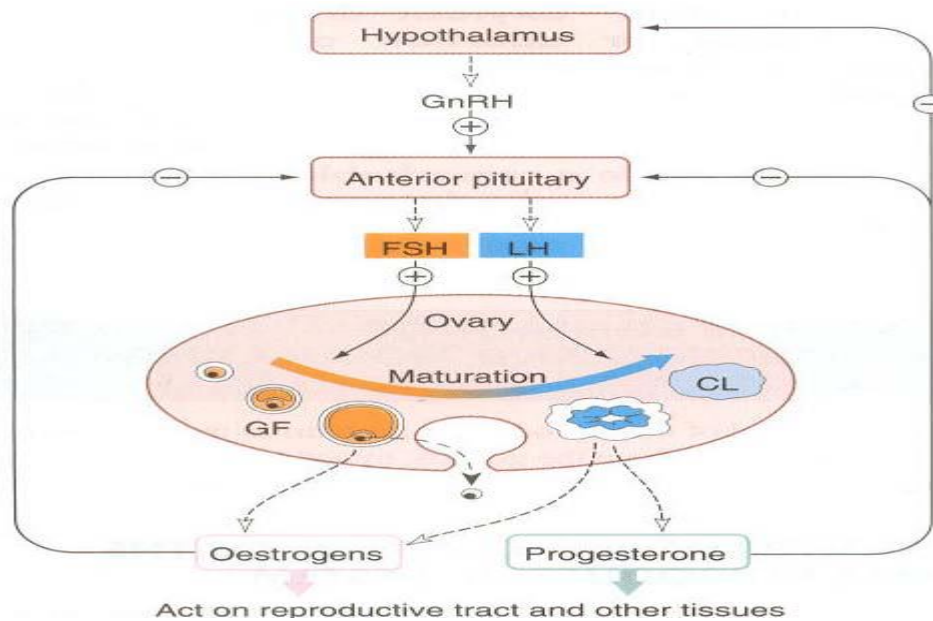


Figure 2. Neuro-hypothalamic-pituitary-ovarian regulation

There is a "synergy of action" between estrogens and progesterone: the action of progesterone can only be exerted on an endometrium already prepared by estrogens.

If there is no fertilization, around the 24th day of the cycle, the corpus luteum atrophies, the production of estrogen and progesterone abruptly stops and the uterine lining peels off: this is menstruation.

If, on the other hand, there is fertilization, the corpus luteum is maintained and with it, the presence of a high level of progesterone: level which will be maintained until an advanced stage of pregnancy, the embryo produces a hormone HCG (chorionic gonadotropin hormone) which ensures the survival of the corpus luteum. The corpus luteum becomes a gestative corpus luteum and continues to secrete estrogen and progesterone for 6 weeks, it will be relayed by the placenta (the pregnancy test is based on the presence of HCG detectable in the urine of the pregnant woman). HCG modifies the immune reaction of the uterus towards the embryo: the uterus behaves as if the embryo were not a foreign body.

4.3. Feedback

Ovarian hormones act in return on the pituitary gland: Feedback.

4.3.1. At the beginning of the cycle

Just before ovulation, FSH triggers follicle maturation and estrogen secretion. When estrogen levels reach a high level, it inhibits FSH production: negative feedback.

4.3.2. At ovulation

A spike in estrogen triggers a sudden surge of LH and FSH leading to ovulation: Positive feedback.

4.3.3. After ovulation

The corpus luteum is formed and increasing amounts of progesterone and estrogen are secreted. When estrogen-progesterone levels increase, it triggers a negative feedback on FSH and LH.

The inhibition of FSH and LH secretion causes the regression of the corpus luteum, the level of ovarian hormones reaches its lowest value; this is the period of menstruation. Due to the collapse of ovarian hormone levels, the secretion of FSH and LH is reawakened and the cycle resumes.

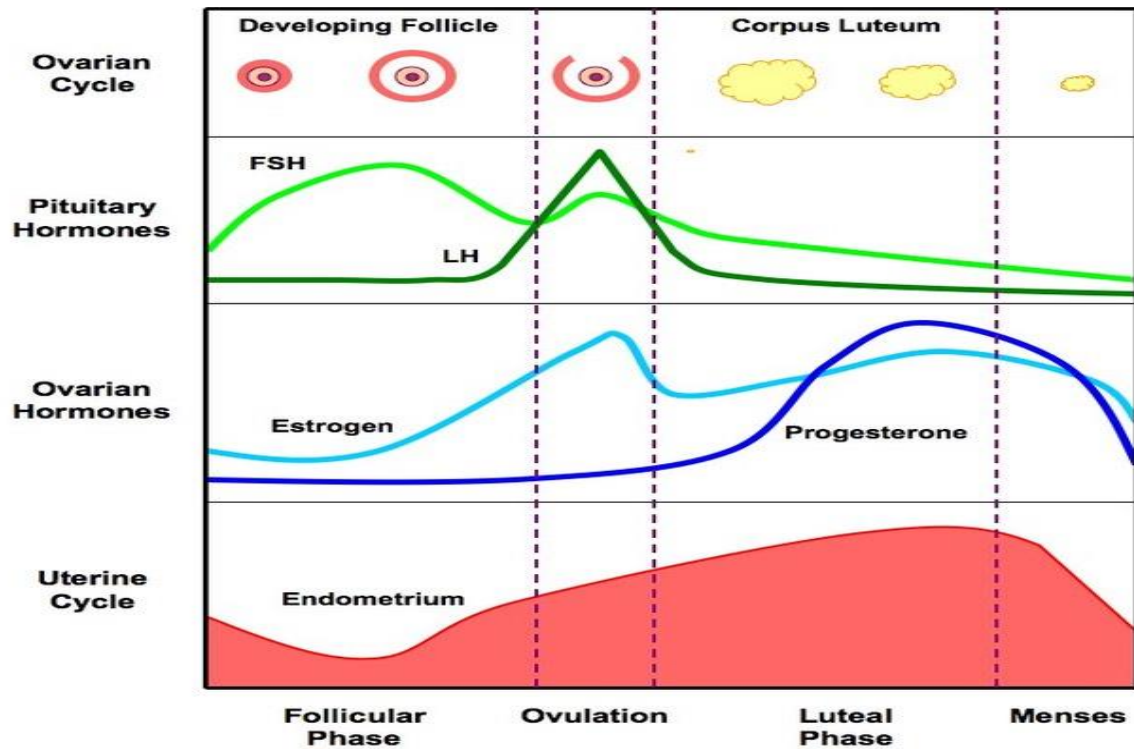


Figure 4. Ovarian cycle, uterine cycle and hormonal variations in women